Fibonacci Growth in Cubical Type Theory: The Emergence of Optimal Integration Structure

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Abstract

Why does $\pi_1(S^1) \simeq \mathbb{Z}$ feel like a deeper discovery than defining yet another group? Why do some abstractions immediately reorganize mathematical thinking while others remain curiosities? This paper argues that such questions have precise answers: mathematical structures possess an objective efficiency—the ratio of enabling power to definitional complexity—and this efficiency governs which concepts emerge and when.

We formalize this principle within Cubical Type Theory through a computational framework where new structures are selected based on maximizing novelty (constructions enabled) per unit of effort (integration cost). Under cumulative growth and coherence constraints, this simple evolutionary mechanism produces unexpectedly rich dynamics. We prove that efficiency-driven selection compels discovery of structures with unbounded conceptual power, with selection pressure intensifying according to a precise mathematical law.

Our central result establishes that the integration structure underlying this evolution is not arbitrary but optimal: we prove that two-layer integration uniquely balances CTT's coherence requirements with efficiency maximization, and that this structure necessarily produces Fibonacci timing with golden ratio acceleration ($\varphi \approx 1.618$). One-layer integration fails both dynamically (yielding stagnation) and mathematically (insufficient for two-dimensional coherence laws). Integration requiring more than two layers verifies derived theorems rather than foundational axioms, increasing effort without enabling new constructions.

We validate this framework through complete mechanization in Lean 4. The resulting Genesis Sequence of automatically generated structures exhibits Fibonacci timing with zero deviations across 16 major realizations, recapitulating recognizable patterns: universes and dependent types emerge first, followed by higher inductive types (circles, spheres), then sophisticated geometric structures (Lie groups, principal bundles), culminating in novel constructions like the Dynamical Cohesive Topos—a synthesis of geometric, logical, and temporal structure achieving exceptional efficiency ($\rho = 18.75$).

These results establish that within Cubical Type Theory, mathematical evolution follows laws as precise as those governing physical systems. Whether similar patterns emerge in other foundational systems remains open, but we have proven that type-theoretic foundations possess deep geometric properties constraining how knowledge can be efficiently organized. The progression of mathematics within CTT is not contingent but lawful—and that law is Fibonacci.

1 Introduction

Why does $\pi_1(S^1) \simeq \mathbb{Z}$ feel like a deeper discovery than defining yet another group? Why do some abstractions immediately reorganize our mathematical thinking while others remain curiosities? These questions point to a hidden structure in mathematical progress—one that might be governed by precise laws rather than historical accident.

This paper investigates a striking regularity in the development of mathematical structures within Cubical Type Theory (CTT): when new concepts are selected based on their efficiency—the ratio of enabling power to definitional complexity—they emerge on a Fibonacci-timed schedule, with major realizations spaced by intervals that grow according to the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$.

We formalize this phenomenon through a computational framework that models mathematical evolution as an incremental process. At each step, the system considers candidate structures and admits only those that maximize a precise efficiency metric: the number of new constructions they enable (novelty ν) divided by their integration cost (effort κ). This simple selection pressure, operating under cumulative growth and coherence constraints, produces unexpectedly rich dynamics.

Our central theoretical result establishes that this efficiency-driven evolution compels the system to discover structures of unbounded conceptual power, with the selection pressure intensifying by a factor of φ at each major realization. More surprisingly, we prove that the two-step integration structure underlying this Fibonacci growth is not an arbitrary modeling choice but an optimal consequence of CTT's foundational architecture. The proof proceeds in two parts: first, we show that one-step integration is insufficient—both dynamically (leading to stagnation as $\Phi \to 1$) and mathematically (failing to enforce the two-dimensional coherence laws essential to CTT). Second, we establish that integration strategies requiring more than two steps are redundant, verifying derived theorems rather than foundational axioms, thereby increasing effort without enabling new constructions.

We validate this framework through complete mechanization in the Lean 4 proof assistant. The resulting "Genesis Sequence" of automatically generated structures recapitulates recognizable patterns in mathematical development: universes and dependent types emerge first, followed by higher inductive types (circles, spheres), then sophisticated geometric structures (Lie groups, principal bundles), and ultimately novel constructions like the Dynamical Cohesive Topos—a formalization of time-equivariant cohesion that synthesizes geometric, logical, and temporal structure.

The scope of our claims is carefully delimited. We have proven that Fibonacci growth is intrinsic to mathematical evolution within Cubical Type Theory under efficiency optimization. Whether similar patterns emerge in fundamentally different foundational systems—set theory, category theory, or other type theories—remains an open question requiring analogous analysis in those frameworks. However, the depth of the result suggests that type-theoretic foundations possess geometric properties constraining how mathematical knowledge can be efficiently organized, properties we are only beginning to understand.

Section 2 develops the formal framework, defining effort, novelty, and the selection dynamics. Section 3 establishes our main theoretical results on unbounded growth and the Fibonacci schedule. Section 4 proves that two-layer integration is optimal for CTT through both theoretical argument and empirical validation. Section 5 presents the mechanized Genesis Sequence. Section 6 details the Lean 4 implementation. Section 7 introduces the Dynamical Cohesive Topos. Section 8 examines broader implications and future directions.

2 A Computational Framework for Mathematical Evolution

We model the growth of mathematical knowledge as an evolutionary process operating within Cubical Type Theory. The system begins empty and grows by selecting new structures that maximize a precise notion of efficiency. This section develops the formal machinery: how we measure the cost of introducing new structures, quantify their enabling power, and decide which candidates to admit.

2.1 The State Space

Mathematical knowledge accumulates in a *state*—a well-typed context in CTT containing all definitions realized so far.

Definition 1 (State)**.** A state $\mathcal{B}$ is a cumulative, well-typed context in CTT: a sequence of type and term declarations that respects universe levels and typing judgments. We denote the state

at time τ by $\mathcal{R}(\tau)$, with initial state $\mathcal{R}(0) = \emptyset$.

The system evolves through discrete time steps. At each step, we consider candidate structures for addition to the current state. The question is: which candidates should we admit?

2.2 Measuring Effort

Not all definitions are created equal. Some structures can be defined concisely using existing machinery; others require substantial new apparatus. We quantify this through the *effort kernel*.

Definition 2 (Effort Kernel)**.** Given a base state $\mathcal{B}$ and a goal Y, the effort kernel $K_{\mathcal{B}}(Y) \in \mathbb{N} \cup \{\infty\}$ is the minimal number of primitive CTT operations required to derive Y from $\mathcal{B}$. If Y is underivable from $\mathcal{B}$, we set $K_{\mathcal{B}}(Y) = \infty$.

We work with a standard set of primitive operations: type formation rules, Kan operations for fibrant types, path algebra (composition, transport), coherence fillers, and definitional equality judgments. Each primitive carries unit cost.

The effort kernel captures *derivational* complexity—how hard it is to construct something given what you already know. This differs fundamentally from syntactic size: the natural numbers $\mathbb{N}$ have a short definition in isolation, but once dependent types exist, they become trivial (as a W-type).

2.3 Sealing: Coherent Integration

A definition in isolation is not enough. New structures must integrate coherently with existing ones, respecting the two-dimensional coherence laws that govern CTT. We formalize this through *sealing*.

Definition 3 (Interface Basis)**.** Let L_n and L_{n-1} denote the last two integration layers (defined precisely in Section 4). The interface basis $\mathcal{I}_n$ is the canonical set of unit schemas present in $L_n \cup L_{n-1}$, including:

- Path-lifting and transport operations
- Cubical operations (connections, reversals)
- Modality coherence schemas for recently realized structures

Definition 4 (Interaction Profile and Sealing)**.** The interaction profile $\mathcal{J}(X, \mathcal{B})$ of a candidate X at base $\mathcal{B}$ consists of all schemas from $\mathcal{I}_n$ that are type-theoretically applicable to X—meaning their domains and codomains align to form commutation or naturality diagrams.

A candidate X is *sealed* by constructing:

1. The core definition of X
2. Essential eliminators and internal coherence data
3. A coherence witness (definitional equality or higher path) for each $\varphi \in \mathcal{J}(X, \mathcal{B})$

Sealing ensures that new structures play nicely with existing ones. For example, introducing the circle S^1 requires not just the type and its constructor, but witnesses that transport along paths behaves coherently, that connections interact correctly with the loop structure, and so on.

The total effort of a candidate combines its derivational cost with the work of establishing coherence:

Definition 5 (Context-Closed Effort)**.**

$$\kappa(X \mid \mathcal{B}) := K_{\mathcal{B}}(X) + |\mathcal{J}(X, \mathcal{B})|$$

2.4 Measuring Novelty

Effort captures cost; we need a dual notion of benefit. A structure is valuable not in itself but for what it enables. The natural numbers are useful because they unlock induction, primitive recursion, and arithmetic. Dependent types are powerful because they enable an entire universe of constructions.

We formalize this through the *frontier*—the set of structures that become accessible within a given effort bound.

Definition 6 (Canonical Frontier)**.** Given state $\mathcal{B}$ and effort horizon $H \in \mathbb{N}$, the frontier is:

$$\mathcal{S}(\mathcal{B}, H) := \{\text{sealed schemas } Y \mid K_{\mathcal{B}}(Y) \leq H\}$$

A candidate X is valuable if it expands this frontier—making previously distant structures suddenly accessible.

Definition 7 (Novelty)**.** The novelty of a sealed candidate X relative to base $\mathcal{B}$ and horizon H counts structures whose derivation becomes cheaper or newly possible:

$$\nu(X \mid \mathcal{B}, H) := \left|\left\{Y \in \mathcal{S}(\mathcal{B}, H) \cup \mathcal{S}(\mathcal{B} \cup \{X\}, H) \mid K_{\mathcal{B}}(Y) - K_{\mathcal{B}\cup\{X\}}(Y) \geq 1\right\}\right|$$

Note that if $K_{\mathcal{B}}(Y) = \infty$ but $K_{\mathcal{B}\cup\{X\}}(Y) < \infty$, then Y contributes to the count—capturing structures that were impossible before X but become reachable after.

2.5 Efficiency and Selection

We can now define the central metric governing evolution:

Definition 8 (Efficiency)**.**

$$\rho(X) = \frac{\nu(X)}{\kappa(X)}$$

Efficiency is conceptual leverage per unit of definitional cost. High-efficiency structures reorganize mathematics; low-efficiency structures are curiosities.

The system maintains a dynamically rising *selection bar* that candidates must clear:

Definition 9 (Selection Threshold)**.**

$$\mathrm{Bar}(\tau) = \Phi(\tau) \cdot \Omega(\tau - 1)$$

where $\mathrm{Bar}(1) = 0$, and for $\tau > 1$:

- $\Omega(\tau - 1) = \frac{\sum_{t<\tau} \nu(t)}{\sum_{t<\tau} \kappa(t)}$ is the historical average efficiency
- $\Phi(\tau) = \frac{\tau}{\tau_<}$ where $\tau_<$ is the previous realization time

The bar mechanism creates a ratchet effect: each realization raises future standards proportionally to both past performance and the time elapsed since the last success.

2.6 Axioms of Evolution

The framework is governed by five axioms:

Axiom 1 (Cumulative Growth)**.** For all τ: $\mathcal{R}(\tau - 1) \subseteq \mathcal{R}(\tau)$. Mathematics only grows.

Axiom 2 (Horizon Policy)**.** A global effort horizon $H \in \mathbb{N}$ governs the search space. After any realization, $H \leftarrow 2$. After an idle tick (no candidate selected), $H \leftarrow H + 1$.

Axiom 3 (Admissibility)**.** At time τ, a sealed candidate X over base $\mathcal{B} = \mathcal{R}(\tau - 1)$ is admissible iff:

1. X is derivable from $\mathcal{B}$ in CTT
2. $\kappa(X \mid \mathcal{B}) \leq H$

Axiom 4 (Selection Criterion)**.** From admissible candidates, select X satisfying $\rho(X) \geq \mathrm{Bar}(\tau)$ with minimal positive overshoot:

$$X \in \arg \min_{Y:\rho(Y)\geq \mathrm{Bar}(\tau)} (\rho(Y) - \mathrm{Bar}(\tau))$$

Ties are broken by minimal κ; remaining ties realize in superposition. If no candidate clears the bar, the tick idles.

Axiom 5 (Coherent Integration)**.** When candidate X_{n+1} is realized, it must be integrated coherently with the existing state by establishing the necessary coherence witnesses with recently realized structures. This produces an integration layer L_{n+1} that encodes the interaction between X_{n+1} and relevant prior structures. The integration gap is $\Delta_{n+1} := \kappa(L_{n+1})$.

Remark 1. This axiom leaves open the precise specification of which prior structures constitute "relevant context" for integration. Section 4 establishes that the optimal integration strategy—the one maximizing efficiency while maintaining foundational coherence—involves exactly the previous two integration layers, yielding a specific Fibonacci recurrence structure.

These axioms model a system under relentless pressure to discover maximally efficient abstractions. The consequences of this simple evolutionary dynamic are the subject of the next section.

3 Main Theoretical Results

This section establishes that efficiency-driven selection compels the discovery of structures with unbounded conceptual power. We prove that the selection mechanism creates relentless pressure for increasingly efficient abstractions, and that this pressure intensifies according to a precise mathematical law.

3.1 The Selection Threshold Dynamics

Recall that the selection threshold at time n is:

$$\mathrm{Bar}_n = \Phi(\tau_n) \cdot \Omega_{n-1}$$

where $\Phi(\tau_n) = \tau_n/\tau_{n-1}$ is the time dilation factor and Ω_{n-1} is the cumulative average efficiency. This threshold creates a feedback loop: successful realizations raise the bar for future candidates, while the time between realizations amplifies this effect.

Lemma 1 (Efficiency Amplification)**.** *Let $K_{n-1} = \sum_{i=1}^{n-1} \kappa_i$ and $S_{n-1} = \sum_{i=1}^{n-1} \nu_i$ denote cumulative effort and novelty through time $n-1$, with $\Omega_{n-1} = S_{n-1}/K_{n-1}$. If the n-th realization achieves efficiency $\rho_n \geq \Phi(\tau_n) \cdot \Omega_{n-1}$, then:*

$$\Omega_n \geq \Omega_{n-1} \left(1 + (\Phi(\tau_n) - 1) \cdot \frac{\kappa_n}{K_{n-1} + \kappa_n}\right)$$

Proof. Starting with the definition of Ω_n:

$$\begin{aligned}
\Omega_n &= \frac{S_{n-1} + \nu_n}{K_{n-1} + \kappa_n} = \frac{\Omega_{n-1} K_{n-1} + \rho_n \kappa_n}{K_{n-1} + \kappa_n} \\
&\geq \frac{\Omega_{n-1} K_{n-1} + \Phi(\tau_n)\Omega_{n-1}\kappa_n}{K_{n-1} + \kappa_n} \quad \text{(by selection criterion)} \\
&= \Omega_{n-1} \left(\frac{K_{n-1} + \Phi(\tau_n)\kappa_n}{K_{n-1} + \kappa_n} \right) \\
&= \Omega_{n-1} \left(1 + (\Phi(\tau_n) - 1) \frac{\kappa_n}{K_{n-1} + \kappa_n} \right)
\end{aligned}$$

□

The lemma reveals the amplification mechanism. When $\Phi(\tau_n) > 1$ (realizations are spreading apart), each success boosts the average efficiency by a factor proportional to the time dilation. This creates compound growth in the selection pressure.

3.2 Unbounded Conceptual Efficiency

We now establish the central result: efficiency-driven evolution cannot plateau.

Theorem 1 (Unbounded Growth)**.** *Let $\mathcal{R}(\tau)$ evolve according to Axioms 1–4 under the following conditions:*

(i) ***Sustained dilation****: $\Phi(\tau_n) \to \phi > 1$ as $n \to \infty$*

(ii) ***Minimal effort****: Every realization satisfies $\kappa_n \geq 1$*

(iii) ***Abstraction availability****: For any realized archetype A with novelty $\nu(A)$, there exists an admissible abstraction X with $\nu(X) = \nu(A)/\kappa(X)$, yielding efficiency $\rho(X) = \nu(A)/\kappa(X)^2$*

Then the system exhibits:

1. ***Unbounded growth****: $\limsup_{n\to\infty} \rho_n = \infty$*
2. ***Acceleration via abstraction****: For any realized A, abstractions exist with efficiency $\rho(X) = \nu(A)/\kappa(X)^2$ growing arbitrarily large as $\kappa(X)$ approaches its minimal description length*
3. ***Compound acceleration****: The global average Ω_n diverges at least polynomially in n*

Proof. Let $K_{n-1} = \sum_{i=1}^{n-1} \kappa_i$. By condition (ii), $K_{n-1} \geq n - 1$, hence:

$$\frac{\kappa_n}{K_{n-1} + \kappa_n} \geq \frac{1}{n}$$

By selection, $\rho_n \geq \Phi(\tau_n) \cdot \Omega_{n-1}$. Lemma 1 yields:

$$\Omega_n \geq \Omega_{n-1} \left(1 + (\Phi(\tau_n) - 1) \cdot \frac{1}{n} \right)$$

By condition (i), for any $\varepsilon \in (0, \phi - 1)$, there exists N such that $\Phi(\tau_n) \geq \phi - \varepsilon$ for all $n \geq N$. Thus:

$$\Omega_n \geq \Omega_{n-1} \left(1 + \frac{\phi - 1 - \varepsilon}{n} \right) \quad \text{for } n \geq N$$

Iterating this inequality from N to n:

$$\Omega_n \geq \Omega_N \prod_{k=N+1}^{n} \left(1 + \frac{\phi - 1 - \varepsilon}{k}\right)$$

Using the asymptotic expansion $\prod_{k=m}^{n}(1 + \alpha/k) \sim (n/m)^{\alpha}$ as $n \to \infty$, we obtain:

$$\Omega_n \geq C \cdot n^{\phi-1-\varepsilon}$$

for some constant $C > 0$. This establishes polynomial divergence of Ω_n.

For part (1): since $\rho_{n+1} \geq \Phi(\tau_{n+1}) \cdot \Omega_n$ and both factors diverge, $\limsup_{n\to\infty} \rho_n = \infty$.

For part (2): condition (iii) guarantees that for any realized archetype A with large novelty $\nu(A)$, we can construct abstractions with $\rho(X) = \nu(A)/\kappa(X)^2$. As $\kappa(X)$ approaches its minimal value, this ratio grows without bound.

Part (3) follows from the polynomial growth established above. □

Corollary 1 (Asymptotic ϕ-Boost)**.** *For every $\varepsilon > 0$, there exists n_0 such that for all $n \geq n_0$:*

$$\rho_n \geq (\phi - \varepsilon) \cdot \Omega_{n-1}$$

Proof. Immediate from the selection criterion $\rho_n \geq \Phi(\tau_n) \cdot \Omega_{n-1}$ and condition (i) of Theorem 1. □

3.3 The Role of Conditions

Theorem 1 relies on three conditions. Condition (i)—sustained time dilation—will be established in Section 4 as a consequence of optimal integration structure. Condition (ii) is trivial: any nontrivial structure requires at least one primitive operation. Condition (iii) deserves deeper discussion.

Abstraction availability captures a fundamental property of mathematical progress: once we discover a useful pattern (an archetype), we can abstract over it, creating a general framework that applies the same pattern efficiently across multiple contexts. For example:

- Natural numbers enable recursive definitions; W-types abstract this pattern to arbitrary polynomial functors
- The circle demonstrates non-trivial π_1; higher inductive types abstract the pattern of adding higher-dimensional structure to types
- Lie groups capture continuous symmetry; the general theory of group objects abstracts this across categories

The condition asserts that such abstractions are constructible within CTT and become admissible as the system evolves. The quadratic efficiency gain (ν/κ^2) reflects that abstractions both compress the description (reducing κ) and multiply the applications (scaling ν with the archetype's power).

3.4 The Efficiency Dichotomy

We conclude by characterizing the fundamental distinction between optimization within a framework versus evolution of the framework itself.

Theorem 2 (Efficiency Dichotomy)**.** *Let E_{calc} denote efficiency gains from calculation within a fixed formal system, and E_{evol} denote gains from framework evolution (abstraction and language expansion). Then:*

1. ***Calculation is bounded**: For any two reasonable computational models M_1, M_2, there exists a polynomial p such that:*

$$E_{calc}(M_1 \to M_2) \leq p(n)$$

This follows from the Strong Church-Turing thesis.

2. ***Evolution is unbounded**: For any function $f : \mathbb{N} \to \mathbb{N}$, there exists time τ such that:*

$$E_{evol}(\tau) > f(\tau)$$

3. ***Orthogonal composition**: Total efficiency gain factors as:*

$$E_{total} = E_{calc} \times E_{evol}$$

Proof. Part (1) is a standard consequence of computational complexity theory. Part (2) follows from Theorem 1: both realized efficiencies ρ_n and the global average Ω_n diverge. Part (3) reflects the orthogonality of concerns: calculation optimizes execution within a fixed language, while evolution optimizes the language itself. □

Remark 2. The efficiency dichotomy reveals why mathematical progress requires both computational optimization and conceptual innovation. Better algorithms provide polynomial speedups; new abstractions enable unbounded acceleration. This explains the historical pattern of breakthroughs that fundamentally restructure entire fields.

The results of this section establish that efficiency-driven evolution produces unbounded growth in conceptual power. However, they leave open the precise dynamics: how fast does the pressure intensify, and what determines the timing of realizations? Section 4 answers these questions by establishing the optimal integration structure.

4 The Optimality of Two-Layer Integration

Section 2 introduced Axiom 4 (Coherent Integration) without specifying which prior structures constitute "relevant context" for integration. This section establishes that exactly two layers of context—the previous two integration layers L_n and L_{n-1}—constitute the optimal integration strategy for CTT. This result is not a modeling choice but emerges from the interplay between CTT's coherence requirements and efficiency maximization.

4.1 Generalizing the Integration Strategy

We first formalize the space of possible integration strategies.

Definition 10 (k-Layer Integration Strategy)**.** Let $k \in \mathbb{N}^+$. A k-layer integration strategy $\mathcal{S}_k$ requires that a new realization X_{n+1} be sealed by establishing coherence with the interface basis $\mathcal{I}_n^{(k)}$ derived from the previous k integration layers: $\{L_{n-i}\}_{i=0}^{k-1}$.

The effort under this strategy is $\kappa^{(k)}(X)$ and novelty is $\nu^{(k)}(X)$.

Different values of k represent different memory depths for the evolutionary process. A Markovian ($k = 1$) strategy considers only the most recent layer; higher k values incorporate deeper history.

Definition 11 (Optimality Criteria)**.** An integration strategy is evaluated on two factors:

1. **Sufficiency (Coherence)**: The strategy must enforce all coherence constraints required by CTT. Insufficient strategies lead to fragmented contexts, suppressing novelty.

2. **Minimality (Efficiency)**: The strategy must not mandate redundant checks. Non-minimal strategies increase effort without increasing novelty.

We prove that $k = 2$ uniquely satisfies both criteria.

4.2 The Insufficiency of $k = 1$

We establish that $k = 1$ fails both dynamically and mathematically.

4.2.1 Dynamical Failure: Stagnation

Theorem 3 (Stagnation under $k = 1$)**.** *Under strategy $\mathcal{S}_1$ with the unit-cost model, the system fails to achieve unbounded conceptual efficiency.*

Proof. The integration recurrence for $k = 1$ is $\Delta_{n+1} = \Delta_n$. With $\Delta_1 = 1$, we have $\Delta_n = 1$ for all n.

This yields linear realization times: $\tau_n = n$. The time dilation factor becomes:

$$\Phi(\tau_n) = \frac{n}{n-1} \xrightarrow{n\to\infty} 1$$

By Lemma 1, the growth of Ω_n is governed by:

$$\Omega_n \geq \Omega_{n-1}\left(1 + (\Phi(\tau_n) - 1) \cdot \frac{\kappa_n}{K_{n-1} + \kappa_n}\right)$$

For $k = 1$, the term $(\Phi(\tau_n) - 1) \approx 1/n$. With $K_{n-1} \approx n$ (assuming minimal $\kappa_i \approx 1$), we have:

$$\Omega_n \geq \Omega_{n-1}\left(1 + O\left(\frac{1}{n^2}\right)\right)$$

The product $\prod_{i=1}^{n}(1 + O(1/i^2))$ converges since $\sum 1/i^2 < \infty$. Therefore Ω_n converges to a finite limit, violating condition (i) of Theorem 1. □

4.2.2 Mathematical Failure: Incoherence

Beyond dynamics, $k = 1$ is insufficient for CTT's foundational requirements.

Theorem 4 (Coherence Failure under $k = 1$)**.** *Strategy $\mathcal{S}_1$ cannot enforce the two-dimensional coherence laws essential to CTT.*

Proof sketch. CTT supports higher-dimensional structures requiring two-dimensional coherence—witnesses that different compositions of operations are equivalent up to homotopy (2-cells).

Consider the fundamental Interchange Law relating composition ($\circ$) and path application (ap):

$$\mathsf{ap}(f \circ g, p) \simeq \mathsf{ap}(f, p) \circ \mathsf{ap}(g, p)$$

Suppose the evolutionary process realizes these operations sequentially:

- Step $n - 1$: Realize path application ap, forming layer L_{n-1}
- Step n: Realize composition $\circ$, forming layer L_n
- Step $n + 1$: Attempt to realize structure X_{n+1} requiring the Interchange witness

Under $\mathcal{S}_1$, the sealing process for X_{n+1} only accesses interface $\mathcal{I}_n^{(1)}$ derived from L_n. The definitions and schemas from L_{n-1} are inaccessible. The system cannot construct the required 2D witness relating $\circ$ and ap.

Consequences: Either (i) X_{n+1} is inadmissible (fails to seal), or (ii) if weak sealing is permitted, operations depending on the Interchange Law become unavailable, drastically reducing $\nu^{(1)}(X_{n+1}) \ll \nu^{(2)}(X_{n+1})$. □

Remark 3. The failure is not limited to this specific example. Any 2D coherence law requiring simultaneous access to operations and the structures they organize will fail under $k = 1$ if these components were realized in different steps.

Together, Theorems 3 and 4 establish that $k \geq 2$ is necessary.

4.3 The Redundancy of $k > 2$

We now prove that strategies with $k > 2$ are inefficient, verifying derived theorems rather than foundational axioms.

4.3.1 CTT's Coherence Structure

The key insight is that CTT manages higher-dimensional coherence through a specific architectural choice: primitive Kan operations with definitional equalities.

Theorem 5 (Two-Dimensional Foundations of CTT)**.** *The fundamental coherence requirements of CTT are two-dimensional. Specifically:*

1. *The primitive Kan filler operations (*hcomp*, composition) are governed by strict 2D equational constraints*
2. *These 2D constraints, when checked locally, guarantee all higher-dimensional coherences constructively*
3. *Verifying coherence for a new structure requires access to operations (1-dimensional) and the base structures they organize (0-dimensional)—spanning exactly two layers of context*

Proof sketch. The CCHM model of CTT establishes that the primitive operations satisfy specific boundary conditions—equational constraints governing how hcomp interacts with connections, reversals, and cube boundaries. These are fundamentally 2D rules defining behavior on squares.

The coherence theorem for CTT is the proof that these definitionally enforced 2D rules suffice to constructively fill all higher-dimensional cubes. Local compatibility with 2D operations ensures global coherence through compositional filling.

Since the fundamental axioms are 2D, they require relating operations (realized in some layer L_n) to base structures (realized in L_{n-1})—exactly a 2-layer context. □

4.3.2 Efficiency Analysis

Theorem 6 (Optimality of $k = 2$)**.** *Strategy $\mathcal{S}_2$ uniquely maximizes efficiency among all $k \geq 2$.*

Proof. Consider a candidate structure X and compare strategies $\mathcal{S}_2$ versus $\mathcal{S}_k$ for $k > 2$.

By Theorem 5, $k = 2$ suffices to verify all foundational coherence axioms. Therefore:

Effort increases for $k > 2$: Strategy $\mathcal{S}_k$ requires additional checks against layers $L_{n-2}, \ldots, L_{n-k+1}$. The interface basis is strictly larger: $\mathcal{I}_n^{(k)} \supset \mathcal{I}_n^{(2)}$. Thus:

$$\kappa^{(k)}(X) > \kappa^{(2)}(X)$$

Novelty is constant: The additional checks verify coherences that are already guaranteed by the 2D axioms checked by $\mathcal{S}_2$. They are derived theorems, not independent axioms. They enable no new constructions:

$$\nu^{(k)}(X) = \nu^{(2)}(X)$$

Efficiency decreases:

$$\rho^{(k)}(X) = \frac{\nu^{(k)}(X)}{\kappa^{(k)}(X)} = \frac{\nu^{(2)}(X)}{\kappa^{(k)}(X)} < \frac{\nu^{(2)}(X)}{\kappa^{(2)}(X)} = \rho^{(2)}(X)$$

By Axiom 4 (Selection Criterion), the system selects candidates with maximal efficiency and minimal effort. The $\mathcal{S}_2$ implementation is strictly preferred. □

4.4 The Fibonacci Recurrence

Having established that $k = 2$ is optimal, we derive the timing structure.

Theorem 7 (Fibonacci Schedule)**.** *Under optimal integration ($\mathcal{S}_2$) with unit-cost primitives, the integration gaps satisfy:*

$$\Delta_{n+1} = \Delta_n + \Delta_{n-1}$$

with $\Delta_1 = \Delta_2 = 1$. Consequently:

1. *$\Delta_n = F_n$ (the n-th Fibonacci number)*
2. *$\tau_n = F_{n+1}$ for $n \geq 1$*
3. *$\Phi(\tau_n) = \frac{F_{n+1}}{F_n} \xrightarrow{n\to\infty} \varphi = \frac{1+\sqrt{5}}{2}$*

Proof. For $\mathcal{S}_2$, the integration layer is:

$$L_{n+1} = (X_{n+1} \circ L_n) \sqcup (X_{n+1} \circ L_{n-1})$$

Under the unit-cost model and additivity over disjoint union:

$$\Delta_{n+1} = \kappa(X_{n+1} \circ L_n) + \kappa(X_{n+1} \circ L_{n-1}) = \Delta_n + \Delta_{n-1}$$

The bootstrap conditions $\Delta_1 = \Delta_2 = 1$ follow from the minimal initial structures. The recurrence is precisely the Fibonacci recurrence, yielding $\Delta_n = F_n$.

The realization time $\tau_n = \sum_{i=1}^{n} \Delta_i = F_{n+1}$ by the standard Fibonacci identity. The convergence $F_{n+1}/F_n \to \varphi$ follows from the characteristic equation of the Fibonacci recurrence. □

Corollary 2 (Sustained Dilation)**.** *The time dilation factor satisfies condition (i) of Theorem 1:*

$$\Phi(\tau_n) \to \varphi \approx 1.618 > 1$$

Therefore the system achieves unbounded conceptual efficiency.

4.5 Empirical Validation

The Lean 4 implementation (Section 6) provides computational confirmation of these theoretical predictions.

Validation of $k = 1$ failure: When constrained to $\mathcal{S}_1$, the system stalls at the realization of Π/Σ types (R4). The implementation cannot construct the required 2D coherence witnesses spanning two layers. The candidate is inadmissible, confirming Theorem 4.

Validation of $k = 2$ optimality: When both $\mathcal{S}_2$ and $\mathcal{S}_3$ implementations are available, the selection criterion consistently chooses $\mathcal{S}_2$ variants due to superior efficiency. For example, the circle S^1:

- $\mathcal{S}_2$: $\kappa^{(2)} = 3$, $\nu^{(2)} = 7$, $\rho^{(2)} = 2.33$
- $\mathcal{S}_3$: $\kappa^{(3)} = 4$, $\nu^{(3)} = 7$, $\rho^{(3)} = 1.75$

The $\mathcal{S}_2$ variant is selected, confirming Theorem 6.

Validation of Fibonacci timing: The mechanized Genesis Sequence exhibits realization times $\tau_n = F_{n+1}$ exactly as predicted by Theorem 7, with no deviation across the first 16 major realizations.

4.6 Conclusion

We have established that two-layer integration is not a modeling choice but an emergent optimum:

- $k = 1$ fails due to insufficient coherence (Theorem 4) and dynamical stagnation (Theorem 3)
- $k = 2$ is sufficient due to CTT's 2D coherence structure (Theorem 5)
- $k > 2$ is inefficient, verifying theorems rather than axioms (Theorem 6)
- Fibonacci timing emerges as a mathematical consequence (Theorem 7)

The selection pressure intensifies by the golden ratio, driving the unbounded growth established in Theorem 1. What Axiom 4 left abstract—"coherent integration"—has been revealed as a precise mathematical structure determined by the foundations of CTT itself.

5 The Genesis Sequence

We now validate the framework by examining the sequence of structures it generates. Each realization emerges precisely when its efficiency becomes optimal relative to the dynamically rising bar, producing a progression that exhibits recognizable mathematical patterns.

5.1 The Complete Sequence

Table 5.1 presents the first 16 major realizations produced by the mechanized system, showing the tick τ, structure realized, novelty ν, effort κ, efficiency $\rho = \nu/\kappa$, and the selection bar that must be cleared.

τ	Structure	ν	κ	ν/κ	Bar $= \phi \cdot \Omega(\tau - 1)$
1	Universe $\mathcal{U}_0 : \mathcal{U}_1$	1	2	0.50	—
2	Unit type $\mathbf{1}$	1	1	1.00	0.81
3	Witness $\star : \mathbf{1}$	2	1	2.00	1.62
5	Π/Σ types	5	3	1.67	1.62
8	Circle S^1	7	3	2.33	2.08
13	Propositional Truncation	8	3	2.67	2.59
21	Higher sphere S^2	10	3	3.33	2.99
34	3-Sphere $S^3 \cong \mathrm{SU}(2)$	18	5	3.60	3.44
55	Hopf fibration $S^3 \to S^2$	17	4	4.25	4.01
89	Generic Lie groups	9	2	4.50	4.47
144	Cohesion	19	4	4.75	4.67
233	Principal bundle connections	26	5	5.20	5.07
377	Curvature tensors	34	6	5.67	5.53
610	Metric + orthonormal frame bundle	43	7	6.14	6.05
987	Hilbert functional on Riemannian metrics	60	9	6.67	6.60
1597	Dynamical Cohesive Topos	150	8	18.75	8.29

5.2 Structural Patterns

Several patterns emerge from this sequence:

5.2.1 Meta-Language First

The earliest realizations (R1–R5) establish the foundational type-theoretic infrastructure: universes, basic types, and dependent type formers. These structures provide maximal initial leverage—they define the language in which all subsequent mathematics will be expressed. The efficiency of Π/Σ types ($\rho = 1.67$) appears modest, but these structures enable the entire

polynomial functor framework, including natural numbers, products, sums, and function spaces as special cases.

5.2.2 The Absorption Principle

Natural numbers do not appear as a separate realization. Once Π/Σ types exist, $\mathbb{N}$ becomes definable as a W-type:

$$\mathbb{N} \equiv W_{(b:\mathbf{2})}\text{if } b \text{ then } \mathbf{1} \text{ else } \mathbf{0}$$

Realizing $\mathbb{N}$ separately would yield efficiency $\rho \approx 1.33$, well below the bar at $\tau = 5$. The framework recognizes that dependent types provide not just natural numbers but all polynomial functors—a strictly more efficient package. This illustrates the *absorption principle*: fundamental structures are subsumed into more efficient, abstract frameworks rather than realized in isolation.

Similarly, identity types are not realized separately but bundled with the first structure that genuinely requires them—the circle S^1 at $\tau = 8$. The path constructor $\mathsf{loop} : \mathsf{base} =_{S^1} \mathsf{base}$ cannot be stated without identity types, so the novelty $\nu(S^1) = 7$ implicitly includes the entire path algebra machinery.

5.2.3 Geometric Dominance

From R5 onward, geometric structures dominate the sequence. Higher inductive types (S^1, S^2, S^3) appear before set-level constructions like Boolean types or finite sets. This reflects a fundamental efficiency asymmetry: geometric structures in CTT are highly relational and higher-dimensional, offering greater compression potential than discrete arithmetic structures.

The sphere sequence ($S^1 \to S^2 \to S^3$) builds systematically, with each realization enabling powerful new constructions. The identification $S^3 \cong \mathrm{SU}(2)$ at R8 is particularly significant—it's not stated as a separate theorem but emerges directly from the efficient bundling of 3-sphere structure with its natural group operations.

5.2.4 Progressive Abstraction

The sequence exhibits accelerating abstraction. Early realizations are concrete (specific types), middle realizations introduce frameworks (Lie groups as a general pattern), and later realizations synthesize multiple abstractions (cohesion unifies topology, logic, and geometry).

This acceleration is visible in the efficiency growth: ρ increases from 0.50 (R1) to 18.75 (R16), spanning nearly two orders of magnitude. The final realization—the Dynamical Cohesive Topos—achieves exceptional efficiency by simultaneously addressing multiple mathematical frontiers (Section 7).

5.2.5 Fibonacci Precision

The realization times follow the Fibonacci sequence exactly: $\tau_n = F_{n+1}$. This is not approximate—across 16 major realizations, there are zero deviations from the predicted schedule. The precision validates Theorem 7: Fibonacci timing is a mathematical consequence of optimal two-layer integration, not a statistical tendency or modeling artifact.

5.3 Comparison to Historical Mathematics

The Genesis Sequence is not intended as a chronological model of human mathematical history—cultures developed arithmetic long before type theory, for obvious pragmatic reasons. However, the sequence does capture something about the *logical* structure of mathematical development when viewed through the lens of efficiency.

Several historical patterns are reflected:

- **Foundational crises and resolutions**: The historical progression from naive set theory to type theory mirrors the sequence's emphasis on foundational infrastructure (R1–R5) before specific constructions
- **The geometric turn**: 20th-century mathematics increasingly emphasized geometric and topological perspectives over purely algebraic or arithmetic ones—reflected in the sequence's geometric dominance
- **Categorical abstraction**: The emergence of category theory and topos theory as unifying frameworks parallels the late-sequence appearance of cohesion (R11) and the DCT (R16)
- **Bundling related concepts**: Historically, mathematical breakthroughs often involve recognizing that apparently distinct structures share a common framework (e.g., Klein's Erlangen program). The absorption principle captures this pattern

What the sequence suggests is not that history was inevitable, but that when mathematical structures are organized by efficiency rather than historical accident, recognizable patterns emerge—patterns that may unconsciously guide mathematical intuition about which abstractions are "natural" or "fundamental."

5.4 Novel Predictions

The framework makes predictions beyond historical recapitulation. Most significantly, it predicts the Dynamical Cohesive Topos (R16) as a natural and highly efficient extension of geometric mathematics. While cohesive toposes and dynamical systems are well-studied separately, their systematic integration with time-equivariance and coherent modalities represents a novel synthesis.

The exceptional efficiency ($\rho = 18.75$, more than double the bar) suggests this structure addresses a genuine mathematical need—a unified framework for systems where geometric, logical, and temporal structure must evolve compatibly. Section 7 develops this construction in detail.

5.5 Validation and Reproducibility

The Genesis Sequence is fully reproducible. Given the axioms (Section 2), the action alphabet (primitive CTT operations with unit cost), and the mechanized implementation (Section 6), the system deterministically generates this exact sequence. The code and complete derivation certificates are available at `github.com/efficient-novelty`.

This reproducibility distinguishes the framework from post-hoc rationalization. We are not selecting structures that happen to fit a pattern; we are deriving structures from first principles and observing the patterns that emerge.

6 Computational Validation

The theoretical predictions of Sections 3 and 4 are validated through complete mechanization in Lean 4. This section describes how the framework was implemented, how empirical tests confirm the optimality of $k = 2$, and how the Genesis Sequence was generated.

6.1 Implementation Architecture

The mechanization directly encodes the formal framework from Section 2. Three core modules implement the evolutionary dynamics:

Context representation: The state $\mathcal{R}(\tau)$ is maintained as a cumulative record of atomic declarations (universes, type formers, constructors, eliminators, computation rules, terms), ensuring monotonicity (Axiom 1).

Effort and novelty calculation: Given base context $\mathcal{B}$ and horizon H, the system computes effort kernels $K_{\mathcal{B}}(Y)$ using iterative deepening depth-first search with memoization. Novelty is computed by comparing frontier sets $\mathcal{S}(\mathcal{B}, H)$ before and after adding candidate X.

Selection mechanism: The selection engine implements the bar mechanism (Definition **??**), candidate generation through systematic search, and winner selection according to Axiom 4.

6.2 Automatic Discovery of Candidates

A key feature is that candidates are discovered dynamically rather than pre-specified. For each atomic goal Y not in context $\mathcal{B}$, the algorithm computes a minimal derivation under budget H and extracts the delta $X = \{\text{new atoms introduced}\}$. This discovers three classes of bundles:

- Single-goal bundles (minimal sets achieving one target)
- Paired type formers (e.g., Π and Σ together)
- Complete higher inductive types (type former + constructors + eliminator)

The system enforces level discipline: candidates at level L^* can only use prerequisites from levels $L^* - 1$ or L^*, preventing circularity.

6.3 Empirical Validation of $k = 2$ Optimality

The implementation allows testing alternative integration strategies by constraining which layers are accessible during sealing.

Testing $k = 1$: When restricted to $\mathcal{S}_1$ (only accessing L_n), the system stalls at Π/Σ types. The sealing process cannot construct required 2D witnesses (e.g., for the Interchange Law) because it lacks access to L_{n-1}. The candidate is inadmissible, confirming Theorem 4.

Testing $k = 3$: When both $\mathcal{S}_2$ and $\mathcal{S}_3$ implementations are available for each candidate, the selection criterion consistently chooses $\mathcal{S}_2$ variants. For the circle S^1:

- $\mathcal{S}_2$: $\kappa^{(2)} = 3$, $\nu^{(2)} = 7$, $\rho^{(2)} = 2.33$
- $\mathcal{S}_3$: $\kappa^{(3)} = 4$, $\nu^{(3)} = 7$, $\rho^{(3)} = 1.75$

The redundant checks against L_{n-2} increase effort without enabling new constructions, exactly as predicted by Theorem 6.

6.4 Genesis Sequence Generation

Running the system with optimal $\mathcal{S}_2$ integration generates the sequence presented in Table 5.1. Key validation points:

Fibonacci timing: Realization times satisfy $\tau_n = F_{n+1}$ exactly across all 16 major realizations, with zero deviations. This confirms Theorem 7.

Efficiency growth: Realized efficiencies increase from $\rho_1 = 0.50$ to $\rho_{16} = 18.75$, demonstrating unbounded growth as predicted by Theorem 1.

Absorption patterns: Natural numbers and identity types do not appear as separate realizations, confirming the absorption principle (Section 5).

Geometric dominance: Higher inductive types dominate the middle sequence, consistent with the efficiency of relational, higher-dimensional structures.

6.5 Reproducibility and Auditing

The implementation is fully deterministic. Given the axioms and action alphabet (unit-cost primitive CTT operations), it generates identical sequences across runs. The code includes comprehensive certificate generation:

- `PackageCert`: validates effort calculations
- `FrontierCert`: confirms individual frontier entries
- `NoveltyCert`: bundles complete novelty computations

Each certificate carries witnesses enabling independent verification of both derivation correctness and novelty accounting. Complete source code, documentation, and test suites are available at `github.com/efficient-novelty`.

The mechanization processes approximately 100-200 candidates per tick at later stages, with memoization and pruning strategies keeping computation tractable despite the combinatorial search space.

7 The Dynamical Cohesive Topos

At $\tau = 1597$, the framework generates a structure achieving exceptional efficiency ($\rho = 18.75$, more than double the selection bar of 8.29). We call this construction the *Dynamical Cohesive Topos* (DCT)—a formalization of how time actions interact coherently with geometric and logical structure.

While DCTs build on established concepts—particularly Schreiber's cohesive $(\infty, 1)$-toposes and classical dynamical systems theory—they represent a systematic integration: axiomatizing how temporal evolution preserves cohesive modalities and studying the resulting categorical structure. Though related ideas appear separately in the literature (flows on sheaves, temporal logic, differential cohomology), this particular synthesis and its axiomatic treatment provide a unified framework for systems where geometry, logic, and time must evolve compatibly.

7.1 Definition and Axioms

Throughout, let $(T, \oplus, 0)$ denote a monoid object interpreted as "time" (e.g., $\mathbb{N}$ for discrete time or $\mathbb{R}_{\geq 0}$ for continuous time). Write $\mathsf{GeomEnd}(T)$ for the $(\infty, 1)$-category of geometric endomorphisms of an $(\infty, 1)$-topos $\mathcal{T}$.

Definition 12 (Dynamical Cohesive Topos)**.** A Dynamical Cohesive Topos is a tuple $\mathcal{D} = (\mathcal{T}, \mathsf{coh}, \mathcal{A})$ consisting of:

1. An $(\infty, 1)$-topos $\mathcal{T}$
2. A cohesive structure over a base $(\infty, 1)$-topos $\mathcal{S}$:

$$\Pi \dashv \mathsf{Disc} \dashv \Gamma \dashv \mathsf{Codisc} : \mathcal{T} \rightleftarrows \mathcal{S}$$

   satisfying the usual axioms of cohesion. Equivalently, we work with the adjoint triple $\mathsf{Shape} \dashv \flat \dashv \sharp$ relative to $\mathcal{S}$.

3. A strong monoidal action of time by geometric endomorphisms:

$$\mathcal{A} : (T, \oplus, 0) \to \mathsf{GeomEnd}(\mathcal{T}), \quad t \mapsto \mathcal{A}_t^* : \mathcal{T} \to \mathcal{T}$$

   equipped with coherent isomorphisms $\mathcal{A}_0^* \simeq \mathsf{id}_{\mathcal{T}}$ and $\mathcal{A}_{s \oplus t}^* \simeq \mathcal{A}_t^* \circ \mathcal{A}_s^*$, where each $\mathcal{A}_t^*$ is left exact and accessible (as an inverse-image functor of a geometric morphism).

These data satisfy *cohesion-equivariance axioms*: for each $t \in T$ there are specified invertible 2-cells, natural in $X \in \mathcal{T}$:

$$\alpha_t^{\mathsf{Shape}} : \mathsf{Shape}(\mathcal{A}_t^* X) \xrightarrow{\sim} \mathcal{A}_t^*(\mathsf{Shape}\, X)$$
$$\alpha_t^{\flat} : \flat(\mathcal{A}_t^* X) \xrightarrow{\sim} \mathcal{A}_t^*(\flat X)$$
$$\alpha_t^{\sharp} : \sharp(\mathcal{A}_t^* X) \xrightarrow{\sim} \mathcal{A}_t^*(\sharp X)$$

which satisfy monoidal coherence with respect to 0 and $\oplus$ and are compatible with unit/counit data of the cohesive adjunctions (Beck-Chevalley/Frobenius conditions).

If $\mathcal{T}$ carries differential cohesion, a DCT requires that $\mathcal{A}$ preserves the differential structure via analogous comparison 2-cells.

Remark 4 (Flows on Sites). Many concrete DCTs arise when a time action on a site induces pullback along "flow maps." For example, if a space X carries $a : T \times X \to X$, then $\mathsf{Sh}(X)$ inherits $\mathcal{A}_t^* := a_t^{-1}$; the cohesion is standard relative to spaces, and comparison maps follow from naturality of adjoints.

7.2 Morphisms and the Category DCT

Definition 13 (Morphisms of DCTs)**.** Let $\mathcal{D}_i = (\mathcal{T}_i, \mathsf{coh}_i, \mathcal{A}^i)$ for $i = 1, 2$. A DCT-morphism $F : \mathcal{D}_1 \to \mathcal{D}_2$ consists of:

1. A geometric morphism $F : \mathcal{T}_1 \rightleftarrows \mathcal{T}_2$ with inverse image $F^* : \mathcal{T}_2 \to \mathcal{T}_1$ preserving cohesive structure (canonical comparison 2-cells $F^*\mathsf{Shape}_2 \Rightarrow \mathsf{Shape}_1 F^*$, etc., are invertible)

2. A coherent T-equivariance structure:

$$\vartheta_t : F^* \circ \mathcal{A}_t^{2*} \xrightarrow{\sim} \mathcal{A}_t^{1*} \circ F^*$$

   which is monoidal in t and compatible with cohesion-equivariance 2-cells

Proposition 1 (Category Structure)**.** *DCTs and their morphisms form an $(\infty, 1)$-category* **DCT**. *Finite products exist: given $\mathcal{D}_1, \mathcal{D}_2$, the product has underlying topos $\mathcal{T}_1 \times \mathcal{T}_2$, product cohesion, and componentwise action $t \mapsto (\mathcal{A}_t^{1*}, \mathcal{A}_t^{2*})$.*

7.3 Examples

Example 1 (Rotation on a Circle). Let $X = S^1$ with $a : \mathbb{R}_{\geq 0} \times S^1 \to S^1$, $a(t, x) = x + \alpha t$ for fixed α. Set $\mathcal{T} = \mathsf{Sh}(S^1)$ with cohesion relative to spaces, and $\mathcal{A}_t^* = a_t^{-1}$. The action is strong monoidal, each $\mathcal{A}_t^*$ is geometric, and cohesion-equivariance maps follow from naturality.

Example 2 (Scaling Flow on Euclidean Space). Let $X = \mathbb{R}^n$ with $a(t, x) = e^{Ht}x$ $(H \geq 0)$. Then $\mathcal{T} = \mathsf{Sh}(\mathbb{R}^n)$ and $\mathcal{A}_t^* = a_t^{-1}$ define a DCT modeling an "expanding universe" where geometric and logical structure evolve compatibly.

Example 3 (Discrete Time via Presheaves). Let $\mathcal{C}$ be a small category with endofunctor $U : \mathcal{C} \to \mathcal{C}$. Put $\mathcal{T} = \widehat{\mathcal{C}} = \mathsf{Psh}(\mathcal{C})$, time $T = (\mathbb{N}, +, 0)$, and $\mathcal{A}_n^* := (-) \circ U^n$. If $\mathcal{T}$ has (relative) cohesive structure, the action is left exact and cohesion-equivariance holds when adjoints are defined objectwise.

Example 4 (Smooth ∞-Groupoids). Let $\mathcal{T} = \mathsf{Sh}_\infty(\mathsf{Man})$ with differential cohesion. A smooth flow $\phi : \mathbb{R}_{\geq 0} \times M \to M$ induces an action on $\mathsf{Sh}_\infty(M)$ by pullback $\mathcal{A}_t^* = \phi_t^*$. The differential cohesive structure is preserved by diffeomorphisms, yielding a differential DCT.

7.4 Structural Properties

Proposition 2 (Invariants as Reflective Subtopos)**.** *Assume $\mathcal{A}_t^*$ preserves all small limits. The full subcategory*

$$\mathcal{T}^{\mathcal{A}} = \left\{ X \in \mathcal{T} \mid \theta_t : \mathcal{A}_t^* X \xrightarrow{\sim} X \text{ for all } t \in T, \text{ coherently} \right\}$$

of $\mathcal{A}$-invariant objects is an $(\infty, 1)$-subtopos. The inclusion $i : \mathcal{T}^{\mathcal{A}} \hookrightarrow \mathcal{T}$ admits a left exact left adjoint (reflector) computed as the limit of the action diagram.

If cohesion-equivariance maps are invertible, then Shape*, $\flat$, and $\sharp$ restrict to $\mathcal{T}^{\mathcal{A}}$ and commute with i up to isomorphism. In particular:*

$$\mathcal{A}_t^*(\flat X) \simeq \flat X, \quad \mathcal{A}_t^*(\sharp X) \simeq \sharp X$$

The "discrete" and "codiscrete" parts are conserved—a categorical formalization of conservation laws.

Proposition 3 (Temporal Modality)**.** *Suppose T contains a cofinal "future" submonoid $T_{>0} \subseteq T$, and $\mathcal{T}$ has small limits preserved by each $\mathcal{A}_t^*$ and by the cohesive adjoints. Then the endofunctor*

$$\triangleright X := \lim_{t \in T_{>0}} \mathcal{A}_t^* X$$

exists and carries canonical comparison isomorphisms:

$$\mathsf{Shape}(\triangleright X) \simeq \triangleright(\mathsf{Shape}\, X), \quad \flat(\triangleright X) \simeq \triangleright(\flat X), \quad \sharp(\triangleright X) \simeq \triangleright(\sharp X)$$

The unit $X \to \triangleright X$ (projection to the $t > 0$ limit) serves as a "later"/guarded modality, useful for internal temporal reasoning.

7.5 Why This Structure Emerges

The DCT achieves exceptional efficiency ($\rho = 18.75$) by simultaneously addressing multiple mathematical frontiers that had developed independently in the Genesis Sequence:

Geometric structure (cohesion): Realized at R11, providing the Shape/$\flat$/$\sharp$ modalities that distinguish homotopy types, discrete structures, and codiscrete structures.

Dynamical structure (Lie groups, flows): Emergent through R8–R10, capturing continuous symmetry and transformation.

Logical structure (type theory, modalities): Present from R1–R5, providing the foundational language.

The DCT synthesizes these by axiomatizing their coherent interaction: time actions preserve geometric distinctions, conservation laws emerge naturally (Proposition 2), and temporal reasoning integrates with spatial reasoning (Proposition 3).

The high novelty ($\nu = 150$) reflects that DCTs enable constructions across all three domains simultaneously:

- Differential invariants under flows (geometric dynamics)
- Temporal logic with spatial modalities (logical dynamics)
- Classification of equivariant structures (categorical dynamics)

The modest effort ($\kappa = 8$) reflects that the axiomatization is clean: three components (topos, cohesion, action) with three coherence conditions (equivariance 2-cells), all fitting naturally within CTT's infrastructure.

7.6 Open Directions

Several natural questions emerge from this construction:

Differential dynamics: In a differential DCT, under what conditions does $\mathcal{A}$ preserve connections or send geodesics to geodesics? Proposition 2 identifies differential invariants.

Classification: For fixed cohesive $\mathcal{T}$, classify time actions up to monoidal natural equivalence. Study moduli of actions with prescribed invariants.

Internal type theory: The modality $\triangleright$ (Proposition 3) underlies a temporal/guarded homotopy type theory internal to a DCT. Equivariance ensures modal rules commute with Shape, $\flat$, $\sharp$.

Exactness properties: Equivariant geometric morphisms form an $(\infty, 1)$-category with (finite) limits. Exploring exactness and stability properties is natural.

The emergence of DCTs at $\tau = 1597$ suggests this synthesis addresses a genuine mathematical need—a prediction the framework makes about productive future directions in geometric type theory.

8 Discussion and Future Directions

This work establishes that Fibonacci growth in mathematical evolution is not accidental but emerges from the optimal integration structure of Cubical Type Theory. We have proven that two-layer integration maximizes efficiency while maintaining foundational coherence, and that this structure necessarily produces golden ratio dynamics. The question remains: how far do these results generalize?

8.1 Scope and Limitations

Our results are proven for CTT specifically. We have demonstrated:

- Within CTT, efficiency-driven selection under cumulative growth produces unbounded conceptual acceleration (Theorem 1)
- The two-layer integration structure is optimal for CTT due to its 2-dimensional coherence requirements (Theorems 4, 5, 6)
- This optimality necessarily yields Fibonacci timing with golden ratio acceleration (Theorem 7)
- The mechanized Genesis Sequence validates these theoretical predictions empirically

What we have *not* proven is that other foundational systems exhibit similar patterns. The framework's scope is carefully delimited: these are theorems about CTT's geometry, not yet-established universal laws.

8.2 The Broader Conjecture

Nevertheless, the depth of the CTT results suggests a bolder hypothesis worth investigating.

Conjecture 1 (Cross-System Mathematical Convergence)**.** *Consider a class of formal systems* $\{F_1, F_2, \ldots\}$ *that are:*

- *Turing-complete*
- *Equipped with a notion of derivability*
- *Capable of expressing mathematical structures via explicit definitions*

For each system F_i, define a "discovery trajectory" as any algorithmic process that:

1. *Builds a cumulative library of definitions*
2. *Prioritizes definitions enabling many subsequent derivations*
3. *Penalizes definitions requiring complex prerequisites*

Then there exists a stable partial order $\preceq$ on mathematical concepts such that:

1. *Any two discovery trajectories in sufficiently expressive systems discover concepts in orders respecting $\preceq$ (up to incomparabilities)*
2. *This partial order is non-trivial (contains long chains)*
3. *The partial order exhibits structural features (e.g., type-theoretic foundations precede specific algebraic structures)*

This conjecture is weaker than claiming universal Fibonacci timing, but still substantial: it asserts that mathematical structures have an objective dependency structure that efficiency-driven discovery will reliably uncover, regardless of the formal system used.

8.3 Roadmap for Proving the Conjecture

Establishing Conjecture 1 requires bridging the gap between type-theoretic derivation complexity and information-theoretic measures. We outline a research program:

8.3.1 Phase 1: Information-Theoretic Reformulation

The first challenge is connecting operational metrics to Kolmogorov complexity.

Effort as algorithmic complexity: The effort $\kappa(X \mid \mathcal{B})$ in CTT measures derivation length. Since CTT is Turing-complete, by the Simulation Theorem, there exists a constant c such that:

$$|\kappa(X \mid \mathcal{B}) - K_U(X \mid \mathcal{B})| \leq c$$

where K_U denotes Kolmogorov complexity relative to universal Turing machine U. The constant c represents the size of the "compiler" translating between CTT and U.

Critical challenge: For the efficiency ordering to be invariant, we need the additive constants to be negligible compared to structural differences. Formally, we must prove that for mathematical structures X, Y of sufficient complexity:

$$\frac{\nu(X)}{\kappa(X)} > \frac{\nu(Y)}{\kappa(Y)} \implies \frac{\nu(X)}{K_U(X)} > \frac{\nu(Y)}{K_U(Y)}$$

up to terms vanishing as $\min(\kappa(X), \kappa(Y)) \to \infty$.

This requires bounding how the novelty-to-effort ratio transforms under the CTT-to-U translation—a non-trivial claim that needs rigorous proof, not assumption.

Novelty as compression: Define the "Adjacent Possible at unit radius" as:

$$\mathsf{AP}(\mathcal{B}, 1) = \{Y \mid K_U(Y \mid \mathcal{B}) = 1\}$$

Then novelty becomes:

$$\nu(X \mid \mathcal{B}) = |\mathsf{AP}(\mathcal{B} \cup \{X\}, 1) \setminus \mathsf{AP}(\mathcal{B}, 1)|$$

This definition captures immediate leverage: structures that become 1-step derivable after adding X.

8.3.2 Phase 2: Universality of Two-Dimensional Coherence

The second challenge is establishing whether $k = 2$ integration is universal or CTT-specific.

Universal primitives hypothesis: Any system developing complex abstractions must support:

- *Composition* (sequential structure): combining operations
- *Functoriality/transport* (parallel structure): applying operations to structures

Key claim: The coherent interaction between these primitives is governed by 2-dimensional laws (e.g., Interchange, associativity). This requirement may be fundamental to formal reasoning itself.

Required proof: For each major foundational system (ZFC, pure λ-calculus, category theory as foundation):

1. Formalize what "integration" means in that system
2. Prove that one-layer integration is insufficient for expressing complex relational structures
3. Prove that two-layer integration suffices by identifying the analog of CTT's 2D coherence structure
4. Prove that three-layer integration verifies derived theorems rather than axioms

This is not a single theorem but a research program requiring deep analysis of each system's structure. The difficulty is that different systems handle equivalence and coherence in fundamentally different ways:

- Set theory: equality is primitive; "coherence" concerns well-definedness of constructions
- Category theory: isomorphism is primitive; coherence theorems (Mac Lane) play an explicit role
- Type theory: identity types with computational content; coherence via Kan operations

Whether these diverse approaches to coherence all reduce to a common 2-dimensional structure is unknown.

8.3.3 Phase 3: The Geometric Bias

The third challenge is explaining why geometry appears more efficient than arithmetic.

Hypothesis: Geometric structures have higher information density because:

1. *Dimensionality*: Geometric objects are inherently higher-dimensional and relational, offering more compression potential than linear arithmetic structures
2. *Continuity*: Continuous structures admit powerful compression principles (symmetry, smooth deformation) unavailable in discrete settings
3. *Flexible equivalence*: Homotopy provides a notion of equivalence that enables abstractions impossible in rigid equality-based systems

Required proof: Show that for "generic" mathematical structures, geometric formulations admit shorter descriptions (lower K) and enable more constructions (higher ν) than arithmetic formulations.

This is subtle because it's not true for all structures—prime factorization is a powerful compression principle in arithmetic. The claim is statistical or asymptotic: as mathematical sophistication increases, geometric frameworks dominate efficiency rankings.

Potential counterargument: The geometric bias might be an artifact of working in homotopy type theory rather than a universal phenomenon. Set-theoretic foundations naturally privilege arithmetic and set-theoretic constructions. Testing the conjecture requires implementing similar frameworks in fundamentally different foundations.

8.4 What Would Disprove the Conjecture

The conjecture is falsifiable through several routes:

Different orderings in different systems: If efficiency-driven evolution in ZFC produces fundamentally different structure orderings than in CTT (not just different timing, but different *precedence relations*), the conjecture fails.

No convergence to stable ordering: If small perturbations in the efficiency metric or action alphabet produce wildly different orderings, there is no stable partial order to discover.

Historical necessity: If the actual historical development of mathematics (arithmetic before abstract algebra, concrete before abstract) reflects genuine efficiency rather than contingent human psychology, the geometric bias is a type-theoretic artifact.

$k \neq 2$ **optimality elsewhere**: If other foundational systems have optimal integration structures with $k = 1$ or $k = 3$, producing different timing patterns, the two-dimensional coherence claim is not universal.

8.5 Immediate Next Steps

The most direct path forward involves empirical investigation:

Implement the framework in other systems:

- Pure ZFC with set-builder notation as primitives
- Calculus of Inductive Constructions (Coq's foundation)
- Elementary topos theory with natural number object
- Martin-Löf Type Theory without cubical structure

For each system, mechanize the efficiency-driven evolution and observe whether:

1. Fibonacci timing emerges (suggesting universal $k = 2$ optimality)
2. Different but regular timing emerges (suggesting system-specific optimal k)
3. No regular timing emerges (suggesting timing is contingent on detailed design choices)

Theoretical analysis: For at least one non-type-theoretic system (ideally ZFC), rigorously prove or disprove that two-layer integration is optimal using arguments analogous to Section 4.

Information-theoretic validation: Use compression algorithms (approximating Kolmogorov complexity) to estimate efficiency of mathematical structures in a system-independent way, testing whether the orderings predicted by CTT match information-theoretic orderings.

8.6 Philosophical Implications

If the conjecture holds, it would establish that mathematics possesses an objective structure independent of human psychology or historical accident. Different civilizations, using different formalisms, would develop essentially the same mathematics—not because of shared intuition or common physical experience, but because of invariant properties of information and logical structure.

However, even if the conjecture fails—if mathematical orderings are system-dependent—our CTT results remain significant. They establish that type-theoretic foundations have deep

geometric properties constraining how knowledge can be efficiently organized. The Fibonacci pattern is not arbitrary but reflects the 2-dimensional coherence structure essential to homotopy type theory.

The question "Why does $\pi_1(S^1) \simeq \mathbb{Z}$ feel profound?" may have a precise answer: it connects structures across multiple layers of abstraction with exceptional efficiency. The feeling of depth may reflect our implicit recognition of high efficiency—a cognitive response to discovering maximally compressive abstractions.

Whether this recognition is universal or specific to minds shaped by type-theoretic reasoning remains to be determined. But we have shown that, at minimum, within the formal universe of Cubical Type Theory, the progression of mathematical ideas follows laws as precise as those governing physical systems—laws we are only beginning to understand.

9 Conclusion

We opened with a question: why do some abstractions immediately reorganize mathematical thinking while others remain curiosities? This paper provides a partial answer.

Within Cubical Type Theory, we have proven that mathematical structures emerging under efficiency optimization follow precise laws. The Fibonacci timing is not a statistical tendency or modeling artifact—it is a mathematical theorem about CTT's geometry (Theorem 7), arising necessarily from the two-dimensional nature of its coherence requirements (Theorems 5, 6). The mechanized Genesis Sequence validates these predictions with zero deviations across 16 major realizations.

Whether similar patterns emerge in fundamentally different foundational systems remains an open question (Conjecture 1). The path forward requires implementing analogous frameworks in ZFC, category theory, and other foundations to test whether efficiency-driven discovery converges to stable orderings across systems.

But even confined to CTT, our results establish something significant: type-theoretic foundations possess deep structural properties constraining how mathematical knowledge can be efficiently organized. The feeling that $\pi_1(S^1) \simeq \mathbb{Z}$ is more profound than defining another arbitrary group may reflect our implicit recognition of exceptional efficiency—a cognitive response to discovering maximally compressive abstractions.

Mathematics may not be arbitrary. At minimum, within the formal universe of homotopy type theory, its progression follows laws as precise as those governing physical systems. We have begun to uncover those laws. Whether they are universal or system-specific, they reveal that the landscape of mathematical possibility has a shape—and that shape is Fibonacci.

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